CTF Preparation Series

CTF Fundamentals (Forensics and OSInt)

Links, slides, challenges links and resources

gianemayumi.notion.site/ctf-series

Forensics

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Forensics

Juan Carlos Ramirez Abadia, who the United States says was one of the most powerful leaders of Colombia's cocaine cartels, was arrested in Brazil last year and last week received a 30-year prison sentence. His wife, a fan of the Japanese cartoon character Hello Kitty, was also jailed.

One room full of pink Hello Kitty items gave an insight into Ramirez Abadia's methods. Brazilian media has said he hid messages about drug operations in e-mail images of the cute cat.

Brazilians throng sale of drug lord's luxuries

Thousands of Brazilians clamoured for a piece of gangster chic on Tuesday, jostling their way into a sale of goods ranging from flat-screen TVs to designer underwear confiscated from a convicted Colombian drug lord.

Reuters / Apr 10, 2008

Forensics

Forensics is the art of recovering the digital trail left on a computer. There are plenty of methods to find data which is seemingly deleted, not stored, or worse, covertly recorded.

Today, we are focusing on:

- General knowledge
- Introduction to Steganography

https://ctf101.org/forensics/overview/

General knowledge

File format

Metadata

General knowledge

File format

Metadata

File extensions

General knowledge

File format

Metadata

File extensions

What if the content of the file does not match the extension?

File Signatures (Magic numbers)

Bytes from a file are used to identify its format.
Generally, 2-4 first bytes of a file.

file

determine file type

```
file <file>
```

xxd

make a hex dump or do the reverse

```
xxd <file>
```

General knowledge

File format

Metadata

File Signatures (Magic numbers)

Bytes from a file are used to identify its format.
Generally, 2-4 first bytes of a file.

```
giane@fedora:~/Downloads$ xxd img.jpg | head -1
00000000: ffd8 ffe0 0010 4a46 4946 0001 0100 0001  ......JFIF......
```

69 66 00 00	ÿØÿá??Exif			
FF D8 FF E0	ÿØÿà	0	jpg	JPEG raw or in the JFIF or Exif file format[17]
			jp2 j2k	

List of file signatures

A file signature is data used to identify or verify the content of a file. Such signatures are also known as magic numbers or magic bytes and are usually inserted at the beginning o...

W Wikipedia

Metadata

Data about data. It can include dates, information, location, comments...

Stamp information is also useful (accessed, created, modified)!

exiftool

Read and write meta information in files

```
exiftool <file>
```

```
giane@fedora:~/Downloads$ exiftool file
ExifTool Version Number         : 13.10
File Name                       : file
Directory                       : .
File Size                       : 8.7 kB
File Modification Date/Time     : 2026:01:29 12:28:15-05:00
File Access Date/Time           : 2026:01:29 12:28:29-05:00
File Inode Change Date/Time     : 2026:01:29 12:28:16-05:00
File Permissions                : -rw-r--r--
Error                           : Unknown file type
```

Steganography

Steganography is the practice of hiding data in plain sight. Steganography is often embedded in images or audio.

Hidden message

Viewable message

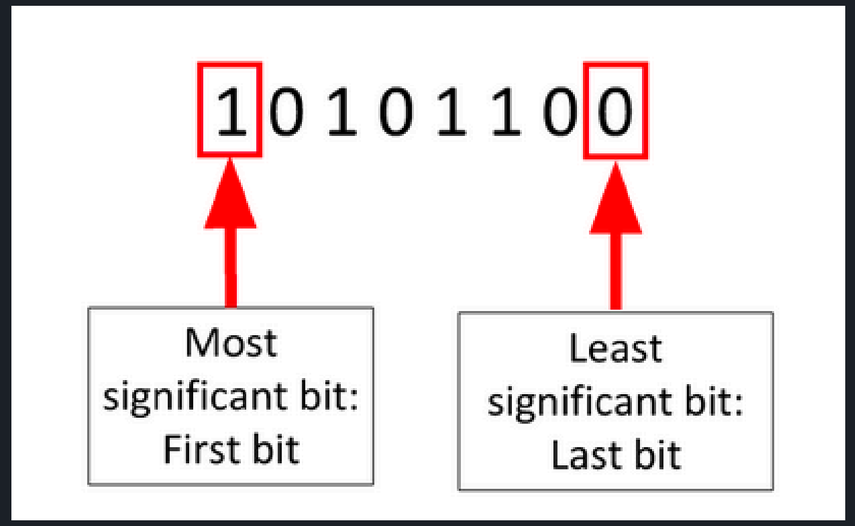
Meet me at the park tonight at 10 pm

Hidden message

Viewable message

Least significant bit (LSB)

The file is composed of bytes. Each one is composed of 8 bits.



Least significant bit (LSB)

Changing the least-significant bit (LSB) doesn't change the value very much.

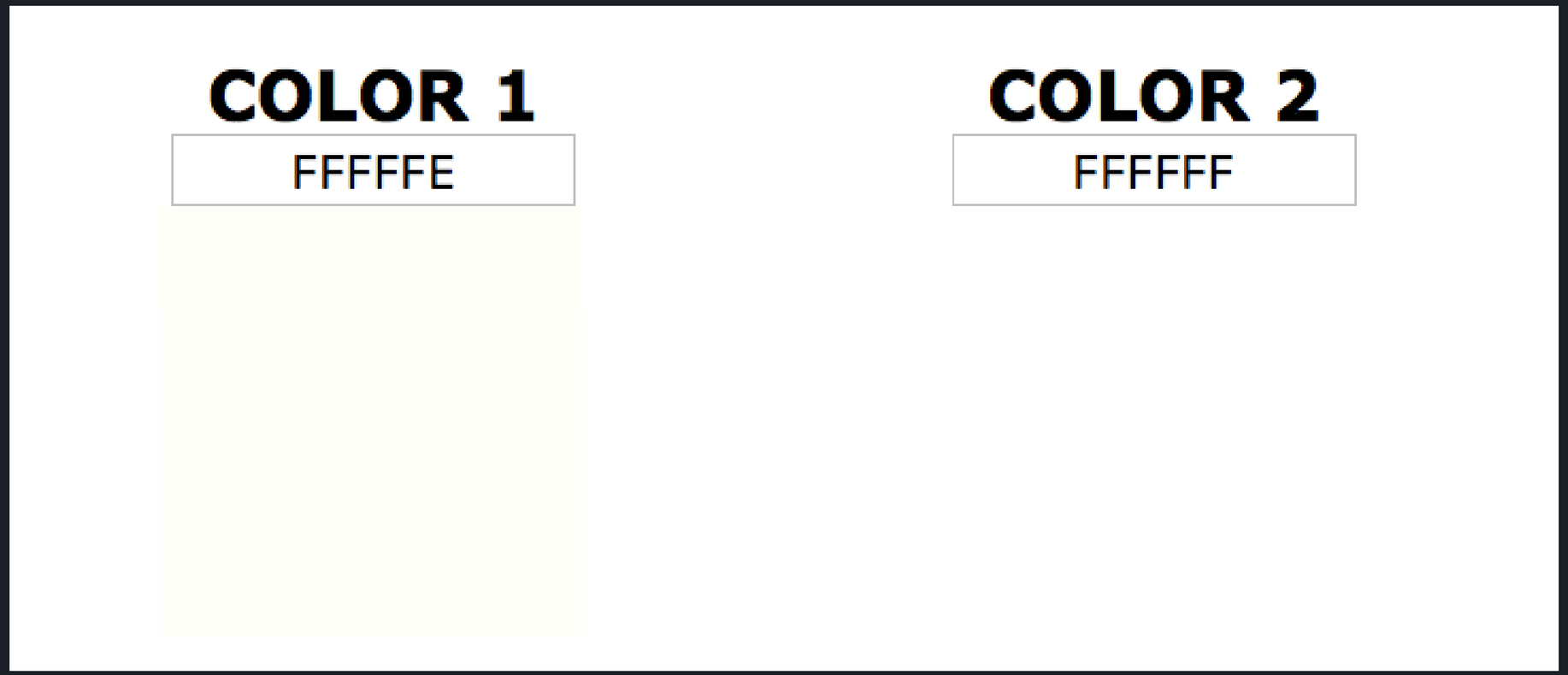
$1\,0\,1\,0\,1\,1\,0\,0_2 = 172_{10}$

changing the LSB from '0' to '1':

$1\,0\,1\,0\,1\,1\,0\,1_2 = 173_{10}$

Changing the LSB from each byte, we can hide a message!

Least significant bit (LSB)



Steganography

steghide

Hide and find data in various kinds of image and video files.

```
steghide --info <file>
```

zsteg

A tool to detect steganography in images, focusing on LSB.

```
zsteg <file>
```

TIP

CyberChef

Open Source Intelligence

The file is composed of bytes. Each one is composed of 8 bits.

Google dorking

A technique that utilizes advanced search operators to uncover information on the internet that may not be readily available through standard search queries.

Google dork cheatsheet

Search filters

Filter	Description	Example
allintext	Searches for occurrences of all the keywords given.	`allintext:"keyword"`
intext	Searches for the occurrences of keywords all at once or one at a time.	`intext:"keyword"`
inurl	Searches for a URL matching one of the keywords.	`inurl:"keyword"`
allinurl	Searches for a URL matching all the keywords in the query.	`allinurl:"keyword"`
intitle	Searches for occurrences of keywords in title all or one.	`intitle:"keyword"`
allintitle	Searches for occurrences of keywords all at a time.	`allintitle:"keyword"`
site	Specifically searches that particular site and lists all the results for that site.	`site:"www.google.com"`

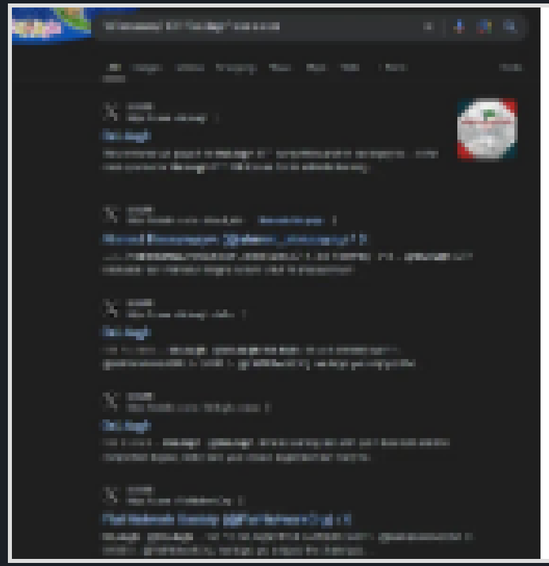
Sherlock

Hunt down social media accounts by username across 400+ social networks

```
giane@fedora:~/Downloads$ sherlock username1
Update available! 0.15.0 --> 0.16.0
https://github.com/sherlock-project/sherlock/releases/tag/v0.16.0
[*] Checking username username1 on:

[+] 1337x: https://www.1337x.to/user/username1/
[+] 7Cups: https://www.7cups.com/@username1
[+] Academia.edu: https://independent.academia.edu/username1
[+] Airliners: https://www.airliners.net/user/username1/profile/photos
[+] AllMyLinks: https://allmylinks.com/username1
[+] AniWorld: https://aniworld.to/user/profil/username1
[+] Anilist: https://anilist.co/user/username1/
[+] Apple Developer: https://developer.apple.com/forums/profile/username1
[+] Apple Discussions: https://discussions.apple.com/profile/username1
[+] Aparat: https://www.aparat.com/username1/
```

Other tools



OSINT CTF Beginner roadmap

Hello, It's Ahmed Elessaway!

Medium / Jan 2

Thank you!

See you next week :)